Remeshing-induced force oscillations and computational cost in a GPU finite-element projection solver

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Abstract

Dynamic mesh adaptation introduces velocity-transfer errors into projection methods and additional work into GPU solvers. We quantify both for two-dimensional incompressible flow with a P1–P1 finite-element solver (SUPG and orthogonal-subscale stabilisation), a BDF2 rotational incremental projection scheme, and newest-vertex bisection with coarsening every 10 ms. After a remesh, both BDF2 velocity levels are transferred and re-projected with the scheme's stabilised pressure operator, a projection-scheme form of the method of Braack, Lang and Taschenberger (2013). In factorial ablations, an unstabilised projection targeting zero discrete divergence produces drag second-difference excursions 300 to 400 times those of the scheme-consistent re-projection at every remesh, whichever velocity levels are re-projected. Retaining the stabilisation contribution in the rotational pressure update makes the pressure error of a manufactured solution grow under time-step refinement. On the DFG 2D-2 and 2D-3 cylinder benchmarks the finest meshes reproduce the maximum drag to within 0.01% and 0.03%. Adaptive refinement reduces the drag error per node at coarse resolution but not the wall time: the finest adaptive run takes 736 s against 259 s for a uniform run of comparable drag accuracy, because of remeshing overhead and twice as many momentum-solver iterations per step.

Keywords: Adaptive mesh refinement, newest-vertex bisection, GPU, projection method, equal-order finite elements, flow past a cylinder, DFG benchmark

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1. Introduction

The laminar flow around a cylinder in a channel defined by Schäfer et al. [1] is the standard test for incompressible solvers. Its periodic case (2D-2, Re = 100) and its transient case (2D-3, inflow ramped up and down over 8 s) have reference values computed with high-order finite elements on very fine meshes and small time steps [2, 3]. Those values are known to four or five significant digits, so a code can be checked against them far more tightly than against the original benchmark intervals.

The flow is localised: a thin boundary layer, two separating shear layers and a vortex street that moves downstream. Local refinement that follows these structures can reduce the number of spatial degrees of freedom needed for a given accuracy, but its net computational benefit is not obvious. It also adds error sources that a fixed mesh does not have. Every remesh transfers the solution to a new mesh; in a projection scheme the transferred velocity is no longer discretely divergence-free, and the history levels that a multi-step time integrator needs must be transferred as well. On a GPU, where the solver itself is fast, the host-side remeshing can dominate the run time. Adaptive finite-element results for the DFG benchmarks are usually goal-oriented and run on CPUs [4, 5, 6], and the effect of the remeshing itself on the forces is rarely isolated. That a velocity moved to a new mesh is no longer discretely divergence-free is well known, and the closest prior work addresses exactly this. Besier and Wollner [7] showed that interpolated or L^2-projected velocities perturb the pressure on the new mesh by $O(1/\Delta t)$, distorting drag and lift, and proposed a discretely divergence-free projection for inf–sup stable elements. Braack et al. [8] extended this to stabilised equal-order elements: the transferred velocity is projected onto the *stabilised* divergence constraint, which contains the stabilisation term of the old pressure; the multiplier of that projection enters the continuity equation of the following time step; the projection is the identity on an unchanged mesh; and the pressure stays bounded independently of Δt for the Stokes equations with backward Euler. Muller et al. [9] transfer all BDF history levels to the new mesh so that an adaptive BDF integrator keeps its order and step size, and study grid-to-grid interpolation and the remeshing frequency for flow past a square cylinder with monolithic Taylor–Hood elements. Di et al.

[10] constructed a divergence-preserving interpolation for moving-mesh finite elements, and Farrell et al. [11], Farrell and Maddison [12] replaced nodal interpolation between unstructured meshes by conservative Galerkin projection on a supermesh. In unfitted finite-element methods on moving domains, the step-to-step change of the discrete problem causes spurious temporal oscillations of the pressure; Olshanskii and von Wahl [13] traced them to a loss of stability of the instantaneous pressure and proposed a variant of the scheme that removes them. On the GPU side, finite-element assembly on unstructured meshes [14, 15] and adaptive finite-element multigrid [16] have been studied, but the cost of frequent remeshing within a time-dependent flow solver, measured against uniform refinement at equal accuracy, has received less attention.

We ask two questions about a solver assembled from these standard components. How large are the force errors that frequent remeshing introduces into a stabilised projection scheme, and which parts of the transfer and of the time discretisation control them? And does adaptivity reduce the wall time on a GPU at the accuracy of the reference data? The contribution is the evidence for the answers, not a new formulation:

1. A pipeline for equal-order P1–P1 elements with SUPG and orthogonal-subscale (OSS) pressure stabilisation, BDF2 rotational incremental projection, and newest-vertex-bisection (NVB) refinement and coarsening at a fixed interval of 10 ms, with assembly and solves on the GPU and the host remeshing overlapped with GPU time stepping (Section 3).
2. Five implementation choices that the verification showed to matter, three for the remeshing and two for the time discretisation, each measured by switching it off (Section 5). The re-projection after a remesh is the projection-scheme counterpart of the stabilised projection of Braack et al. [8], applied to both BDF2 levels as Muller et al. [9] transfer their histories; we write out its equations, state where it differs from theirs, measure how far it is from the identity on an unchanged mesh, and separate by a factorial ablation the effect of the history level from that of the projection operator. For the rotational pressure term we derive what our variant omits.
3. A verification against the DFG reference data: a fully specified manufactured solution, spatial and temporal convergence on uniform and adaptive meshes, 2D-3, and the effect of the outlet condition, which differs from both references (Section 4).

4. A cost account on one GPU that separates the gain per degree of freedom from the gain in wall time (Section 6). The adaptive meshes are modestly ahead per node at the coarse end only, and not ahead in wall time. We give the breakdown that explains why.

2. Problem and reference quantities

The incompressible Navier–Stokes equations

$$\partial_t \mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} - \nu\Delta\mathbf{u} + \nabla p = \mathbf{0}, \qquad \nabla\cdot\mathbf{u} = 0 \tag{1}$$

are solved in the channel $\Omega = (0, 2.2)\times(0, 0.41)\,\mathrm{m}$ minus a cylinder of diameter $D = 0.1\,\mathrm{m}$ centred at $(0.2, 0.2)$, with $\nu = 10^{-3}\,\mathrm{m}^2/\mathrm{s}$; p is the kinematic pressure (pressure divided by the constant density), and all equations below are written for it. The inflow is the parabola $u(0, y) = 4U(t)\,y(H - y)/H^2$, $H = 0.41\,\mathrm{m}$, and no-slip holds on the walls and the cylinder.

The two references treat the outlet differently. FEATFLOW, the 2D-2 reference, uses the natural (do-nothing) outflow condition of its weak formulation [3]; John [2], the 2D-3 reference, prescribes the inflow parabola at the outlet as well. Our default is the usual choice for a projection scheme: the pressure increment vanishes at the outlet, and the momentum predictor has the natural condition $\nu\,\partial_n\mathbf{u} = \mathbf{0}$ there. Because the rotational update of Section 3.2 acts at the outlet nodes too, the outlet pressure is not exactly zero but carries the accumulated rotational correction; at $t = 8\,\mathrm{s}$ of 2D-2 it is below 7×10^{-4} in magnitude on U4 and below 0.016 on A2 (the pressure difference across the cylinder is about 2.5). Neither reference condition is therefore matched exactly. Section 4.5 repeats runs with the parabola prescribed at the outlet and measures the effect.

- **2D-2:** $U = 1.5\,\mathrm{m/s}$, mean velocity $\bar{U} = 1\,\mathrm{m/s}$, $\mathrm{Re} = \bar{U}D/\nu = 100$. The flow is run from rest to $t = 8\,\mathrm{s}$; statistics are taken over the last full periods in $t \geq 6.5\,\mathrm{s}$ (Section 4.2 checks the remaining transient against runs to $t = 30\,\mathrm{s}$). Reported are $C_{D,\max}$, $C_{D,\min}$, $C_{L,\max}$, $C_{L,\min}$, the Strouhal number $\mathrm{St} = fD/\bar{U}$ from the lift period, and the pressure difference $\Delta p = p(0.15, 0.2) - p(0.25, 0.2)$ at half a period after the lift maximum, all with the definitions of the FEATFLOW reference data [3].

- **2D-3:** $U(t) = 1.5\sin(\pi t/8)$ for $0 \leq t \leq 8\,\mathrm{s}$, starting from rest. Reported are $C_{D,\max}$, $C_{L,\max}$, the times at which they occur, and $\Delta p(8\,\mathrm{s})$, against John [2].

Drag and lift coefficients are $C_D = 2F_D/(\bar{U}^2 D)$ and $C_L = 2F_L/(\bar{U}^2 D)$, with the forces per unit density and per unit depth and $\bar{U} = 1\,\mathrm{m/s}$ in both cases. The reference values are listed in Tables 3 and 5.

3. Method

3.1. Spatial discretisation

Velocity and pressure are continuous piecewise-linear on triangles. Convection is written in skew-symmetric form, $(\mathbf{w}\cdot\nabla\mathbf{u},\mathbf{v}) + \frac{1}{2}((\nabla\cdot\mathbf{w})\mathbf{u},\mathbf{v})$, so the Galerkin convection term does not inject energy when the discrete advecting velocity $\mathbf{w}$ is not exactly solenoidal [17]. Streamline-upwind Petrov–Galerkin (SUPG) stabilisation [18] adds the test function $\tau_K\,\mathbf{w}\cdot\nabla\mathbf{v}$ to the mass, convection and pressure-gradient terms of the momentum predictor. The parameter is

$$\tau_K = \left[\left(\frac{2}{T}\right)^2 + \left(\frac{2|\mathbf{w}|}{h_K}\right)^2 + 9\left(\frac{4\nu}{h_K^2}\right)^2\right]^{-1/2}, \qquad h_K = \sqrt{2|K|}, \tag{2}$$

in the transient form of Tezduyar and Osawa [19]. Usually $T = \Delta t$. We fix $T = 1\,\mathrm{ms}$ instead, so that the spatial stabilisation does not change when Δt is refined. With $T = \Delta t$ the SUPG term perturbs the BDF2 time derivative by $O(\Delta t)$, and the observed temporal order fell to one. The same inconsistency of a Δt-dependent τ was analysed by Jia and Esmaily [20], who replace Δt by a physical time scale computed every step from the ratio of the L^2 norms of velocity and acceleration. Our constant T is a fixed, flow-independent approximation to such a time scale, and it is one more tuned constant; Section 5 measures the sensitivity to it. The transient term itself is needed: without it this fractional-step scheme with a lagged pressure diverged on coarse cells (see Bochev et al. [21] for the stability of transient SUPG). Section 5 measures how much the results depend on T.

Equal-order elements in a projection scheme need pressure stabilisation. We use orthogonal subscales [22, 23]: the pressure Poisson operator is augmented by

$$S(q,r) = \sum_K s_K\,(\nabla q, \nabla r)_K - \sum_i \frac{1}{m_i}\,G_i(q)\cdot G_i(r), \tag{3}$$

where

$$G_i(q) = \sum_{K\ni i} \sqrt{s_K}\,\frac{|K|}{3}\,\nabla q|_K,$$

$s_K = \tau_K^p/\Delta t$, $\tau_K^p = c_{\text{oss}}\,(4\nu/h_K^2 + 2|\mathbf{u}^*|/h_K)^{-1}$, and m_i the lumped mass of node i. S penalises only the part of $\sqrt{s}\,\nabla p$ that is orthogonal to its lumped L^2 projection onto continuous P1 functions, so smooth pressures are not penalised. It is symmetric positive semi-definite. We treat the projection implicitly, which keeps the operator SPD; a lagged projection over-damped the flow. The factor $c_{\text{oss}} = 0.3$ was set by the wall-pressure test of Section 5. With $c_{\text{oss}} = 0.03$ a node-to-node pressure mode appeared on the cylinder.

3.2. Time discretisation

The incremental pressure-correction scheme in rotational form [24, 25] is advanced with BDF2. With $\mathbf{w} = 2\mathbf{u}^n - \mathbf{u}^{n-1}$ and $\Delta t' = 2\Delta t/3$, one step is:

1. momentum predictor for $\mathbf{u}^*$ (both components with the same matrix):

$$\frac{3\mathbf{u}^* - 4\mathbf{u}^n + \mathbf{u}^{n-1}}{2\Delta t} + (\mathbf{w}\cdot\nabla)\mathbf{u}^* - \nu\Delta\mathbf{u}^* + \nabla p^n = \mathbf{0},$$

   with SUPG and the Dirichlet data of time t^{n+1};
2. pressure increment δp from the stabilised Poisson problem

$$(\nabla\delta p, \nabla q) + S(p^n + \delta p, q) = -\frac{1}{\Delta t'}(\nabla\cdot\mathbf{u}^*, q);$$

3. velocity correction $\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t'\, M_L^{-1} G\,\delta p$ with the lumped mass matrix M_L;
4. rotational pressure update $p^{n+1} = p^n + \delta p - \nu\, d^*$, where d^* is a nodal approximation of $\nabla\cdot\mathbf{u}^*$ discussed next.

The first step is backward Euler.

The divergence in the rotational term. Our d^* is not the discrete divergence of $\mathbf{u}^*$, and the difference matters. Let K be the P1 stiffness matrix, S the matrix of (3), and D the discrete divergence, $(D\mathbf{u})_i = (\nabla\cdot\mathbf{u}, N_i)$. On every row without a pressure Dirichlet condition, step 2 reads $(K+S)\,\delta p = -(1/\Delta t')\,D\mathbf{u}^* - Sp^n$, so the lumped discrete divergence of $\mathbf{u}^*$ is exactly

$$M_L^{-1} D\mathbf{u}^* = -\Delta t'\, M_L^{-1}\left[K\,\delta p + S\,(p^n + \delta p)\right]. \tag{4}$$

In the continuous scheme $S = 0$ and $\nabla\cdot\mathbf{u}^* = \Delta t'\,\Delta\delta p$. We use the first term only,

$$d^* = -\Delta t'\, M_L^{-1} K\,\delta p,$$

that is, the rotational correction is applied to the Galerkin part of the divergence and not to the part that the stabilisation carries. Compared with $d^* = M_L^{-1} D\mathbf{u}^*$, the update therefore omits the term

$$\nu\, \Delta t'\, M_L^{-1} S\, (p^n + \delta p). \tag{5}$$

This is a modification of the scheme, not an identity. We have no stability proof for either variant, and we keep the modification because of the observed behaviour, which the following algebra helps to interpret. (i) The two variants differ only where $S(p^n + \delta p, \cdot) \neq 0$, that is, only by the part of the pressure gradient that is orthogonal to its continuous projection; for a pressure that S does not penalise they coincide. (ii) If kept, (5) is added to the pressure explicitly at every step. S is symmetric positive semi-definite, so $M_L^{-1}S$ has non-negative eigenvalues, and taken in isolation the increment amplifies the node-to-node modes that S is meant to suppress; since $s_K = \tau_K^p/\Delta t$, the factor $\nu\Delta t' s_K = \frac{2}{3}\nu\tau_K^p$ does not depend on Δt, while the number of steps grows like $1/\Delta t$. This is not a stability analysis: in the coupled scheme the pressure increment, the stabilisation and the velocity correction interact. (iii) What we observe is consistent with it. On U3 at $t = 0.1\,\mathrm{s}$ with $\Delta t = 1$, 0.5 and 0.25 ms (fixed mesh, so this is a pure time-step refinement), the L^2 pressure error of the manufactured solution with our d^* is 1.10, 1.01 and 0.97×10^{-3}, whereas with $d^* = M_L^{-1} D\mathbf{u}^*$ it is 0.97, 2.84 and 8.56×10^{-2}: it grows by about three each time Δt is halved, and the velocity error grows with it. Omitting (5) prevents this growth in the present discretisation. Section 5 shows the effect on the benchmark.

3.3. Forces

Drag and lift are evaluated as volume integrals [26, 2]:

$$F_D = -\left[(\partial_t \mathbf{u}, \mathbf{v}_d) + (\mathbf{u}\cdot\nabla\mathbf{u}, \mathbf{v}_d) + \nu(\nabla\mathbf{u}, \nabla\mathbf{v}_d) - (p, \nabla\cdot\mathbf{v}_d)\right], \tag{6}$$

with $\mathbf{v}_d = (\chi, 0)$, where χ is the P1 function equal to one on the cylinder nodes and zero elsewhere ($\mathbf{v}_l = (0, \chi)$ for lift). The time derivative is the scheme's own BDF2 difference. Only the layer of elements touching the cylinder contributes. Surface-traction forces are computed as well, but only as a cross-check. Δp is the difference of the P1 pressure interpolated at the two probe points.

3.4. Adaptive mesh

Base mesh and bisection. The base mesh has 508 nodes and 896 triangles: an O-grid in the square $[0, 0.41]^2$ around the cylinder and a structured block downstream, with every quadrilateral split along a diagonal. Both triangles of a quadrilateral get the diagonal as their refinement edge. This satisfies Stevenson's matching condition, so newest-vertex bisection produces conforming meshes with a bounded set of shapes, and the conformity closure only bisects coarser elements [27, 28, 29]. One refinement level is two bisections (it halves h). Midpoints of cylinder edges are projected onto the circle. Elements are kept in a forest; coarsening removes newest vertices whose bisection patch is entirely marked for coarsening, at most one level per adaptation. Every adapted mesh is checked (conformity, orientation, boundary tags) before it is used. The minimum angle of the base mesh is 15.29°; no adapted mesh in this paper went below 15.19°.

Indicators. Both indicators measure the disturbance of the undisturbed channel flow. The inflow parabola $u_P(y)$ has the same shear and curvature at every x; an indicator that sees it refines the whole channel. We therefore subtract the nodal interpolant of u_P (scaled with the current inflow in 2D-3) from u_h before recovering a continuous gradient G' by area-weighted averaging [30, 31]. On element K,

$$\eta_{v,K} = h_K \max_{i \in K} \left| \omega_i - \omega_P(y_i) \right|, \tag{7}$$

$$\eta_{i,K} = \max_{\text{edges } e=(i,j)} \tfrac{1}{8} \left| (G'_j - G'_i)\, \mathbf{e} \right|, \tag{8}$$

where ω is the recovered vorticity. η_v is the velocity jump across the element caused by rotation and shear. It sees the boundary layer, the shear layers and the shed vortices. η_i estimates the P1 interpolation error of the disturbance velocity. It sees the stagnation and acceleration region in front of the cylinder, which carries little vorticity but sets the surface pressure. In an early version of the code that used the vorticity indicator alone, that region stayed coarse and $C_{L,\max}$ came out 8% low.

Marking. Each indicator is divided by its tolerance, $\hat{\eta}_v = \eta_v/\mathrm{tol}_v$ and $\hat{\eta}_i = \eta_i/\mathrm{tol}_i$. Elements with $\max(\hat{\eta}_v, \hat{\eta}_i) > 1$ are refined, largest first and within an element budget. Each element is refined by as many levels as its worst indicator needs, using $\eta_v \sim h$ and $\eta_i \sim h^2$. An element is coarsened only if its merged parent is predicted to stay below 0.35 on both indicators. The

band between 0.35 and 1 is a hysteresis that prevents refine–coarsen cycling. Four further rules complete the marking:

- *look-ahead*: indicators above tolerance are traced downstream with the local velocity over one adaptation interval and imposed along the path, so that shed vortices move into mesh that is already refined;
- *buffer and hole filling*: one layer of vertex neighbours is refined with each marked element, and an element coarser than two of its edge neighbours is refined;
- *minimum level*: no element is coarser than level 2. Because the indicators ignore the undisturbed parabola, the approach flow would otherwise stay on the base mesh;
- *keep radius*: no element whose centroid is within 0.25 m of the cylinder centre is coarsened.

In 2D-3 the tolerances are multiplied by $\max(|U(t)|/1.5, 0.1)$. With fixed tolerances the mesh coarsened wholesale as the inflow died away, and remesh spikes contaminated $\Delta p(8\,\mathrm{s})$.

The mesh is adapted every 10 steps (10 ms at $\Delta t = 1$ ms; the interval is scaled with Δt so that it stays 10 ms). Table 1 lists the three adaptive configurations used throughout, and Fig. 1 shows an A2 mesh.

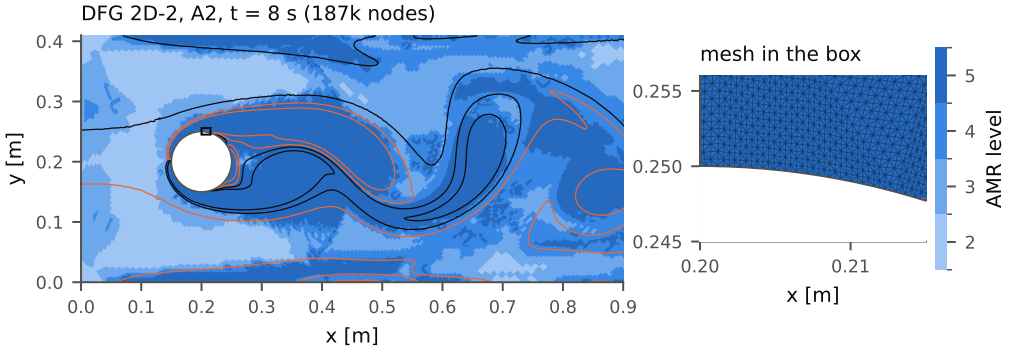


Figure 1: DFG 2D-2, mesh A2 at $t = 8$ s: refinement level with vorticity contours ($\omega = \pm 5, \pm 20\,\mathrm{s}^{-1}$), and the mesh in the small box on top of the cylinder.

Table 1: Mesh configurations. Uniform meshes UL refine the base mesh L levels. Adaptive meshes A1–A3 use the tolerances and maximum level shown, minimum level 2, adaptation every 10 ms. Node counts are means over $t \geq 6.5$ s of DFG 2D-2.

mesh	tol_v [m/s]	tol_i [m/s]	max level	element budget	nodes (2D-2)
U3	–	–	3	–	29 152
U4	–	–	4	–	115 648
U5	–	–	5	–	460 672
A1	0.1	0.01	4	200 000	50 764
A2	0.05	0.005	5	400 000	187 229
A3	0.025	0.0025	6	1 500 000	707 574

Transfer and re-projection. On refinement, new midpoint values are interpolated linearly; on coarsening the values at the remaining vertices are kept (injection). This transfer is neither conservative nor divergence-preserving; conservative Galerkin projection on a supermesh [11, 12] or a divergence-preserving interpolation [10] are the principled alternatives, and we did not try them. Both $\mathbf{u}^n$ and $\mathbf{u}^{n-1}$ are transferred, so the BDF2 scheme keeps its history, as in Muller et al. [9]. The transferred fields do not satisfy the discrete constraint of the scheme on the new mesh, and we re-project both levels with the scheme's own pressure step. To make precise what this does, let G be the discrete gradient, $(Gq)_i = (\nabla q, N_i)$. In the implementation the velocity correction acts only on nodes without a velocity Dirichlet condition (interior and outlet nodes), so G and M_L in the correction are restricted to those nodes; D acts on all nodal velocities, and the Dirichlet values on the inlet, the walls and the cylinder enter it as data; the pressure equations are taken on all rows except the outlet rows, where $\delta p = 0$. Midpoints created on the inlet, the walls and the cylinder during refinement receive the exact Dirichlet values, so the transfer preserves the boundary data. With these restricted operators, $E = K + DM_L^{-1}G$ is the defect between the stiffness matrix and the Schur complement of the lumped projection. Steps 2 and 3 of a time step give, with $p_*^n = p^n + \delta p$ the pressure before the rotational update,

$$D\mathbf{u}^{n+1} + \Delta t'\, S\, p_*^n = -\Delta t'\, E\, \delta p, \tag{9}$$

so $\mathbf{u}^{n+1}$ satisfies the stabilised divergence constraint, not $D\mathbf{u} = 0$, up to the lumping defect. After a remesh, each transferred field $\tilde{\mathbf{u}}$ is re-projected with

the current pressure p, which is not changed:

$$(K+S)\,\phi = -\frac{1}{\Delta t'}\,D\tilde{\mathbf{u}} - S\,p, \qquad \mathbf{u} = \tilde{\mathbf{u}} - \Delta t'\,M_L^{-1}G\,\phi, \tag{10}$$

hence

$$D\mathbf{u} + \Delta t'\,S\,(p+\phi) = -\Delta t'\,E\,\phi. \tag{11}$$

This is the projection-scheme form of the stabilised projection of Braack et al. [8], their Eqs. (12)–(13): with $q = \Delta t'\,\phi$ as the multiplier, the old-pressure term of their constraint is our $\Delta t'\,Sp$ and their Darcy stabilisation S_h^D is our S. It differs from theirs in three respects. (i) It is an approximate projection: the mass matrix is lumped and the Darcy Schur complement $-DM_L^{-1}G$ is replaced by K, which leaves the defect E. (ii) It is applied to both BDF2 levels, each with the current pressure $p = p^n$. (iii) By default the multiplier is discarded, whereas Braack et al. [8] add $-S_h^D(q_h,\cdot)$ to the continuity equation of the next time step (their Eq. (21)); the analogue here is to use $S(p^n+\phi)$ instead of Sp^n in the next pressure step. We have no counterpart of their pressure bound.

Because of (i) and (ii) the re-projection is not exactly the identity on an unchanged mesh, unlike their Lemmas 3 and 5. Inserting (9) into (10) for $\tilde{\mathbf{u}} = \mathbf{u}^n$ on the same mesh gives

$$(K+S)\,\phi = E\,\delta p - S\,(p^n - p_*^n),$$

where $p^n - p_*^n$ is the rotational increment of step 4; in addition the weights s_K are re-evaluated with the current velocity. For $\mathbf{u}^{n-1}$ the right-hand side also contains S applied to the pressure change over one step. All these terms are small, and Section 5 measures them by applying the re-projection every 10 steps on an unchanged mesh.

The comparator in Section 5, which an earlier version of the code applied to $\mathbf{u}^n$ alone, is the unstabilised projection

$$K\,\phi = -\frac{1}{\Delta t}\,D\tilde{\mathbf{u}}, \qquad \mathbf{u} = \tilde{\mathbf{u}} - \Delta t\,M_L^{-1}G\,\phi, \qquad \text{hence} \qquad D\mathbf{u} = -\Delta t\,E\,\phi, \tag{12}$$

with the same restricted operators and the same pressure condition at the outlet. It targets zero discrete divergence but, because the Schur complement is again replaced by K, reaches it only up to the lumping defect; it is not an exact projection onto $D\mathbf{u} = 0$. It is far from the identity, because by (9) it removes the stabilised part of the divergence that the time step leaves. With it, the volume-integral drag shows a spike at every remesh (Fig. 6).

Overlapped remeshing. At a remesh step n the indicators and the state are downloaded, and a host thread runs the marking, refinement, coarsening, mesh extraction, CSR graph construction and validation. The GPU meanwhile advances $K = 5$ more steps on the old mesh. At step $n+K$ the state of that step is transferred to the new mesh with the same rules that the remesh would have used, and the re-projection follows. The step at which the new mesh is used is fixed, so the results do not depend on thread timing. The look-ahead horizon grows by K steps. A unit test checks that the transfer at $n + K$ reproduces the direct transfer bit for bit when the state has not changed. In DFG 2D-2 the overlap changed the errors by at most 0.015 percentage points, and in 2D-3 on A3 by at most 0.007 percentage points apart from $\Delta p(8\,\mathrm{s})$ (Sections 5 and 7).

3.5. GPU implementation and linear solvers

The code is written in CUDA C++ and uses double precision throughout. Element assembly uses one thread per element with atomic additions into CSR matrices whose pattern is built once per mesh. The momentum system (one matrix for both components) is solved by BiCGSTAB [32] with a multicolour ILU(0) preconditioner [33], both components in one solve, to a relative residual of 10^{-7}. The pressure system is solved by conjugate gradients with a smoothed-aggregation algebraic multigrid preconditioner [34] to a relative residual of 10^{-7}. The OSS projection term of (3) is applied inside CG as $-B^{\mathsf{T}} M_L^{-1} B$ with an assembled B. The AMG setup (strength threshold 0.08, greedy aggregation, one damped Jacobi smoothing step of the tentative prolongator, Galerkin coarse operators, dense inverse below 300 unknowns) runs on the host with OpenMP after every remesh and at least every 50 steps; the symmetric V(2,2) cycle with damped Jacobi smoothing runs on the GPU. The remeshing runs on the host (8 OpenMP threads). The new mesh, the CSR pattern and the ILU colouring are then uploaded.

All runs in this paper used one NVIDIA RTX 5080 (16 GB) with an AMD Ryzen 9 9900X host, CUDA 13.3 and gcc 13.3. Every timed run started only when no other process used the GPU, and a watcher logged any other GPU process during the run. One ablation run overlapped another job for about 30 s; its errors are reported but its wall time is not.

4. Verification

4.1. Manufactured solution

The manufactured solution is defined on the benchmark domain, with the cylinder, by a stream function:

$$\psi = \frac{A(t)}{b}\, f(x)\, g(y), \qquad u = \partial_y \psi, \qquad v = -\partial_x \psi,$$
$$p = \frac{A(t)}{2}\, \sin\big(k(L-x)\big)\cos(by), \qquad A(t) = 1 + \tfrac{1}{2}\sin(10\pi t),$$
$$f(x) = \cos\big(k(x-L)\big) - \tfrac{1}{4}\cos\big(2k(x-L)\big), \qquad g(y) = \sin(by) + \tfrac{1}{2}\cos(by),$$
$$k = 2\pi/0.82\,\mathrm{m}^{-1}, \qquad b = \pi/H, \qquad L = 2.2\,\mathrm{m}.$$

The velocity is exactly solenoidal. Since $f'(L) = f''(L) = 0$ and $p(L, y) = 0$, it satisfies the outlet conditions of the code exactly ($\delta p = 0$ and $\nu \partial_n \mathbf{u} = \mathbf{0}$), and the outlet also fixes the pressure level, so no gauge correction is needed. The exact velocity is prescribed at the inlet, the walls and the cylinder nodes. The body force $\mathbf{f} = \partial_t \mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} - \nu\Delta\mathbf{u} + \nabla p$ ($\nu = 10^{-3}$) is evaluated at the nodes at t^{n+1} and integrated as a P1 function with the consistent mass matrix plus its SUPG weight. The run starts from the exact solution at $t = 0$. The uniform meshes U1–U5 are run to $t = 0.1\,\mathrm{s}$ with $\Delta t \propto h$ ($\Delta t = 1\,\mathrm{ms}$ on U1), so this is a coupled space–time refinement: the observed orders combine the spatial and temporal errors. Errors are absolute, measured at $t = 0.1\,\mathrm{s}$ on the polygonal computational domain with a seven-point, degree-5 quadrature rule per triangle; $\max_\Gamma$ is the largest nodal pressure error on the cylinder. Table 2 lists them. The velocity converges at the optimal orders, two in L^2 and one in H^1. The pressure converges at between 1.5 and 2. Rotational projection with $\Delta t \propto h$ predicts $O(\Delta t^{3/2})$ for the pressure [25]. The meshes in this test are fixed. Applying the remesh re-projection every 10 steps on these fixed meshes leaves the errors unchanged to three significant digits (Section 5), but the transfer between different meshes is verified only through the benchmark convergence of the adaptive series and the ablations.

4.2. DFG 2D-2: spatial convergence

All 2D-2 runs use $\Delta t = 1\,\mathrm{ms}$ and run to $t = 8\,\mathrm{s}$.

Periodic state. The FEATFLOW reference values are taken from a fully developed flow (25–30 s), whereas our statistics average the two lift periods in $6.5 \le t \le 8\,\mathrm{s}$. The flow is not yet exactly periodic there: on the finest meshes

Table 2: Manufactured solution on uniform meshes, $\Delta t \propto h$, $t = 0.1\,\mathrm{s}$. Observed orders in brackets.

level	nodes	Δt [s]	$\|\mathbf{u}-\mathbf{u}_h\|_{L^2}$	$\|\mathbf{u}-\mathbf{u}_h\|_{H^1}$	$\|p-p_h\|_{L^2}$	$\max_\Gamma \|p-p_h\|$
1	1 912	1.00×10^{-3}	1.57×10^{-2}	1.45×10^{0}	6.52×10^{-2}	8.46×10^{-2}
2	7 408	5.00×10^{-4}	4.24×10^{-3} (1.89)	5.98×10^{-1} (1.28)	4.42×10^{-3} (3.88)	8.01×10^{-3} (3.40)
3	29 152	2.50×10^{-4}	8.08×10^{-4} (2.39)	2.85×10^{-1} (1.07)	9.71×10^{-4} (2.19)	1.85×10^{-3} (2.12)
4	115 648	1.25×10^{-4}	1.68×10^{-4} (2.27)	1.42×10^{-1} (1.00)	1.79×10^{-4} (2.44)	4.31×10^{-4} (2.10)
5	460 672	6.25×10^{-5}	4.28×10^{-5} (1.97)	7.09×10^{-2} (1.00)	5.37×10^{-5} (1.73)	1.39×10^{-4} (1.63)

the two periods differ by 0.0024% (U5) and 0.0025% (A3) in $C_{L,\max}$, and by at most 0.001% in $C_{D,\max}$, St and Δp. To measure the remaining transient, U4 and A1 were run to $t = 30\,\mathrm{s}$. On U4 the statistics over $28.5 \le t \le 30\,\mathrm{s}$ differ from those over $6.5 \le t \le 8\,\mathrm{s}$ by 0.004 percentage points in $C_{L,\max}$ and $C_{L,\min}$ and by at most 0.0003 in $C_{D,\max}$, St and Δp, and they no longer change after $t = 10\,\mathrm{s}$. On A1 the differences are at most 0.003 percentage points. On these two meshes the 8-s statistics are therefore biased by up to about 0.004 percentage points in the lift. The other meshes were not run to $t = 30\,\mathrm{s}$, so this is a measured bias on U4 and A1 and the close agreement of the two periods on U5 and A3, not a rigorous bound for every mesh; we quote it alongside the comparisons below. Table 3 lists the errors against FEATFLOW level 6, and Fig. 2 plots them against the mean number of nodes. On the uniform meshes $C_{D,\max}$ and $C_{L,\max}$ converge at observed orders 1.98 and 2.14; St converges at order 1.27. Richardson extrapolation of the uniform series gives $C_{D,\max} = 3.22854$ and $C_{L,\max} = 0.98848$. The adaptive series gives orders 1.77 and 1.93 and extrapolates to 3.22787 and 0.98883. Both $C_{L,\max}$ extrapolations lie about 0.2% above FEATFLOW. Section 4.3 shows that this is the time error at $\Delta t = 1\,\mathrm{ms}$. Δp does not converge monotonically on either series. It stays within +0.05% to +0.22% of the reference; we return to it in Section 7.

4.3. DFG 2D-2: time step

On one adaptive and one uniform mesh, A2 and U4, Δt was halved twice (2, 1 and 0.5 ms; on A2 the adaptation interval stays 10 ms). Table 4 shows second-order convergence in time of the force maxima on both. On A2 the successive differences of $C_{D,\max}$ and $C_{L,\max}$ fall by a factor of 4.16 (observed order 2.06), on U4 by 3.97 and 3.96. Extrapolated at the observed order, the time error at $\Delta t = 1\,\mathrm{ms}$ is +0.028% in $C_{D,\max}$ and +0.185% in $C_{L,\max}$ on A2, and +0.030% and +0.201% on U4. The time error is thus nearly the same

Table 3: DFG 2D-2, $\Delta t = 1\,\mathrm{ms}$: relative error [%] against FEATFLOW level 6 [3], mean nodes over $t \geq 6.5\,\mathrm{s}$, and wall time of the whole run ($t = 0$–$8\,\mathrm{s}$, one run each on an otherwise idle GPU).

mesh	nodes	$C_{D,\max}$	$C_{D,\min}$	$C_{L,\max}$	$C_{L,\min}$	St	Δp	wall [s]
FEATFLOW lv6		3.22739	3.16426	0.98658	-1.02129	0.30184	2.48520	
U3	29 152	−0.511	−0.458	−1.469	+1.737	+0.093	+0.403	32
U4	115 648	−0.104	−0.099	−0.188	+0.260	+0.046	+0.175	63
U5	460 672	−0.000	−0.005	+0.106	−0.082	+0.027	+0.217	259
A1	50 764	−0.203	−0.196	−0.308	+0.350	−0.059	+0.060	64
A2	187 229	−0.054	−0.058	+0.076	−0.065	−0.039	+0.197	166
A3	707 574	−0.006	−0.015	+0.186	−0.172	−0.015	+0.057	736
FEATFLOW lv5		−0.024	−0.024	−0.039	+0.035	+0.007	+0.008	

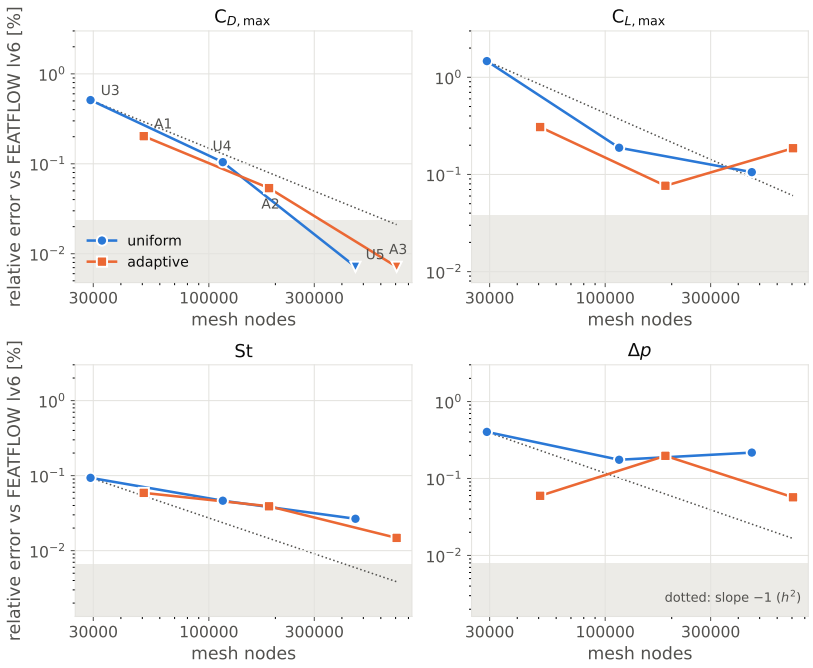


Figure 2: DFG 2D-2, $\Delta t = 1\,\mathrm{ms}$: absolute relative error against FEATFLOW level 6 versus the mean number of nodes, for uniform (U3–U5) and adaptive (A1–A3) meshes. Dotted guide: slope -1 (error $\propto h^2$). Grey band: difference between FEATFLOW levels 5 and 6. Triangles mark errors below the plotted range. $C_{L,\max}$ includes the time error of about 0.2% at this time step (Section 4.3).

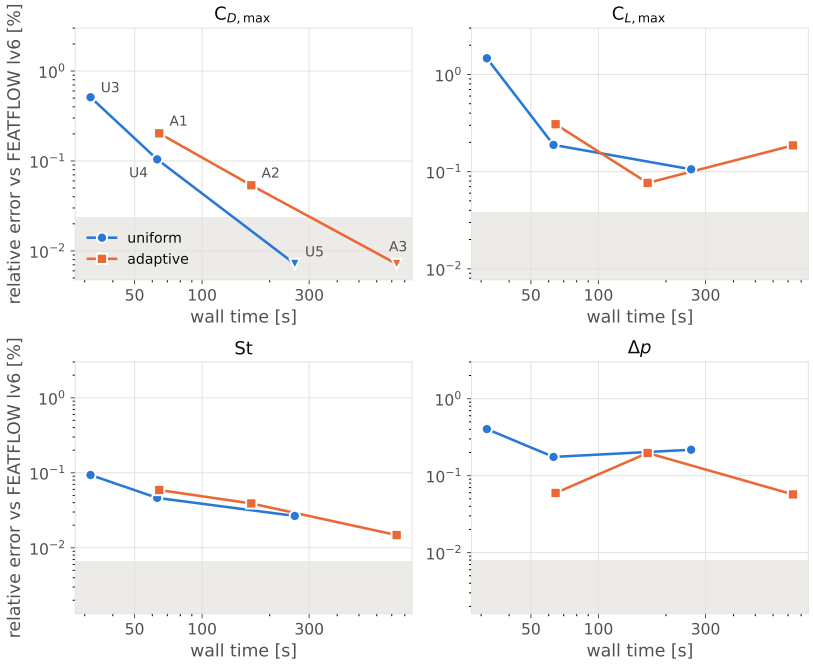


Figure 3: DFG 2D-2, $\Delta t = 1\,\mathrm{ms}$: the errors of Fig. 2 versus the wall time of the whole run ($t = 0$–$8\,\mathrm{s}$, one run per mesh on an otherwise idle GPU).

on the two meshes, which differ in node count by 60% and in mesh type; the two estimates differ by 0.002 and 0.016 percentage points. Δp does not show a clean order on A2 (ratio 2.11, with changes below 0.04%), although it does on U4 (4.07); we do not claim a temporal order for it. This time error, not a spatial error, is why the directly computed $C_{L,\max}$ of the finest meshes at $\Delta t = 1\,\mathrm{ms}$ is +0.11% (U5) and +0.19% (A3) above FEATFLOW, and why both spatial extrapolations of $C_{L,\max}$ lie about 0.2% above it.

The following space–time values are estimates, not computed results. They rest on the assumption, supported by the two meshes above but not proven, that the time error at $\Delta t = 1\,\mathrm{ms}$ is independent of the mesh. Subtracting from each spatial extrapolation the time error measured on a mesh of the same family gives $C_{D,\max} = 3.22758$ and $C_{L,\max} = 0.98650$ for the uniform series (+0.006% and −0.009% against FEATFLOW level 6), and 3.22698 and 0.98700 for the adaptive series (−0.013% and +0.043%). Using the other family's time error instead changes these estimates by at most 0.016 percentage points. For comparison, FEATFLOW level 5 differs from level 6 by −0.024% and −0.039% in these two quantities; this indicates the resolution of the reference data, not a rigorous error bound. The estimates also carry the periodic-state bias measured on U4 and A1, up to 0.004 percentage points in the lift (Section 4.2); the outlet condition changes them by less than 0.001 (Section 4.5).

Table 4: DFG 2D-2, meshes A2 and U4: time-step study. Values and differences between successive Δt.

mesh	Δt [ms]	$C_{D,\max}$	$C_{L,\max}$	St	Δp
A2	2	3.22848	0.99311	0.30192	2.49092
A2	1	3.22566	0.98733	0.30172	2.49010
A2	0.5	3.22498	0.98595	0.30168	2.48971
	difference 2 → 1	-2.82×10^{-3}	-5.78×10^{-3}	-2.02×10^{-4}	-8.26×10^{-4}
	difference 1 → 0.5	-6.79×10^{-4}	-1.39×10^{-3}	-3.86×10^{-5}	-3.91×10^{-4}
	ratio	4.16	4.16	5.22	2.11
U4	2	3.22687	0.99059	0.30216	2.49123
U4	1	3.22402	0.98472	0.30198	2.48955
U4	0.5	3.22331	0.98324	0.30193	2.48914
	difference 2 → 1	-2.84×10^{-3}	-5.87×10^{-3}	-1.80×10^{-4}	-1.68×10^{-3}
	difference 1 → 0.5	-7.16×10^{-4}	-1.48×10^{-3}	-4.77×10^{-5}	-4.13×10^{-4}
	ratio	3.97	3.96	3.78	4.07

4.4. DFG 2D-3

Table 5 compares the 2D-3 results with John [2], and Fig. 4 shows the force histories. On A3, $C_{D,\max}$ is -0.027% from the reference and its time is correct to 10^{-5} relative. $C_{L,\max}$ carries a time error at $\Delta t = 1\,\mathrm{ms}$. On U4 the values at $\Delta t = 1$, 0.5 and 0.25 ms converge at second order in time: the successive differences fall by a factor of 3.98, and the extrapolated value 0.47638 is -0.33% from the reference. On A2 the same refinement gives a ratio of 5.5, not a clean order; its value at $\Delta t = 0.25\,\mathrm{ms}$ is -0.28% from the reference. At $\Delta t = 1\,\mathrm{ms}$ the time error of about $+0.4\%$ therefore hides an offset of about -0.3% that is not a time error, and it is not caused by the outlet condition either (Section 4.5). The A3 value at 1 ms is 0.19 percentage points above A2, so part of the offset is spatial; we did not run A3 at a smaller time step. $C_{D,\max}$ and the times of both maxima do not depend on Δt to within 0.015%. $\Delta p(8\,\mathrm{s})$ is small (-0.11), flat in Δt on U4 (-0.36 to -0.38%), and sensitive to the mesh history of the adaptive runs (Section 7).

Table 5: DFG 2D-3: relative error [%] against John [2]; $t(\cdot)$ is the time of the maximum. Meshes marked * prescribe the parabola at the outlet, as John [2] does (Section 4.5); the others use our default outlet condition.

mesh	Δt [ms]	$C_{D,\max}$	$t(C_{D,\max})$	$C_{L,\max}$	$t(C_{L,\max})$	$\Delta p(8\,\mathrm{s})$	wall [s]
John (2004)		2.95092	3.93625	0.47795	5.69313	−0.1116	
U3	1	−0.386	−0.027	−0.704	+0.053	−1.259	29
U4	1	−0.101	−0.004	+0.134	−0.010	−0.357	59
A2	1	−0.085	−0.003	+0.073	+0.030	−0.400	184
A3	1	−0.027	−0.001	+0.267	+0.012	−0.198	892
U4	0.5	−0.101	−0.005	−0.212	−0.005	−0.376	123
U4	0.25	−0.101	−0.005	−0.300	−0.004	−0.381	266
A2	0.5	−0.086	−0.015	−0.224	+0.036	−0.418	319
A2	0.25	−0.086	−0.005	−0.278	+0.037	−0.459	616
U4*	1	−0.101	−0.004	+0.134	−0.010	−0.357	60
U4*	0.5	−0.101	−0.005	−0.213	−0.005	−0.376	126
U4*	0.25	−0.101	−0.005	−0.300	−0.004	−0.381	271
A2*	1	−0.086	−0.002	+0.090	+0.030	−0.315	193

4.5. Outlet condition

To measure the effect of the outlet condition (Section 2), we added the option of prescribing the inflow parabola at the outlet, as John [2] does. The

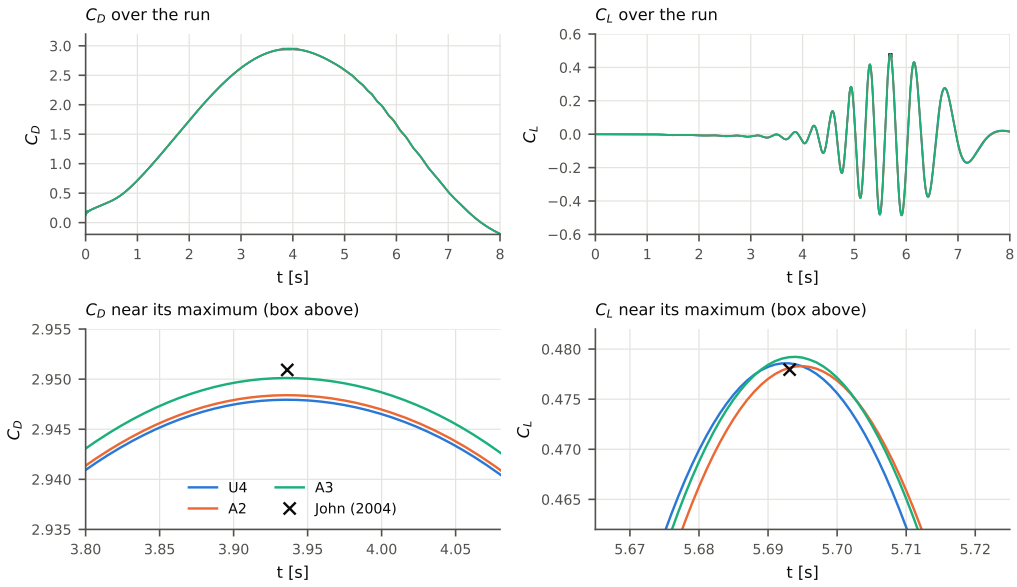


Figure 4: DFG 2D-3, $\Delta t = 1\,\mathrm{ms}$: drag and lift histories on U4, A2 and A3 (top) and near the maxima (bottom), with the reference maxima of John [2].

pressure is then determined only up to a constant; the pressure increment is fixed at the outlet node nearest to mid-height, after the right-hand side of the pressure equation has been made orthogonal to the constants. On an adaptive mesh the inlet and outlet carry different nodes, and the trapezoidal fluxes of the two interpolated parabolas then differ at $O(h^2)$, which makes the pressure problem incompatible; we therefore scale the outlet parabola so that the two discrete fluxes are equal. After a remesh the transferred outlet velocities of both time levels are rescaled in the same way, so that the reprojection also sees a compatible flux; without this, every coarsening of the outlet edges produced a one-step drag spike.

The effect on the forces is negligible (Tables 6 and 5). On U4 the 2D-2 quantities change by at most 0.0003 percentage points and the 2D-3 quantities, at all three time steps, by at most 0.001; the time-extrapolated 2D-3 $C_{L,\max}$ is -0.329% from John [2] with either condition, so its offset is not caused by the outlet condition. On A1 the 2D-2 quantities change by at most 0.0004 percentage points. On the 2D-3 A2 mesh $C_{D,\max}$ changes by 0.001 and $C_{L,\max}$ by 0.017 percentage points, and $\Delta p(8\,\mathrm{s})$ by 0.09, all within the mesh-history noise of adaptive runs (Section 7). The outlet lies 20 di-

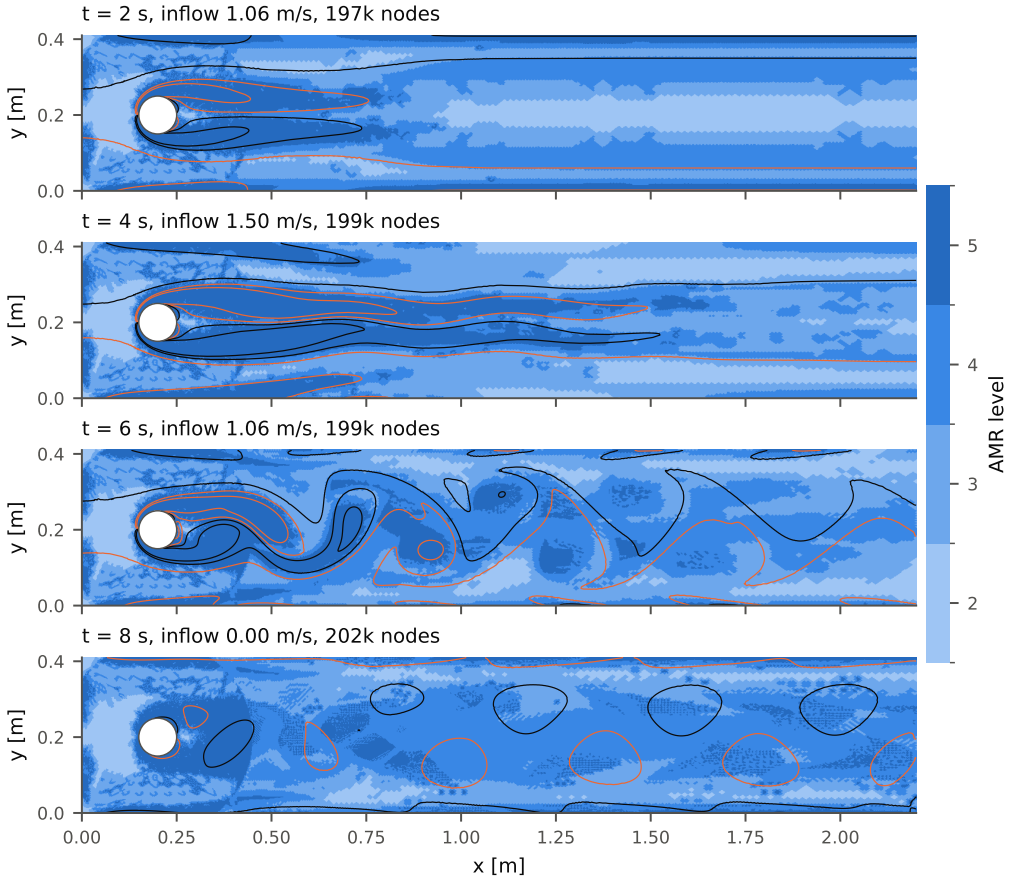


Figure 5: DFG 2D-3, mesh A2: refinement level and vorticity contours ($\omega = \pm 5, \pm 20\,\mathrm{s}^{-1}$) during the inflow cycle. The mesh follows the growing wake and coarsens again as the inflow dies away.

ameters downstream, and at $\mathrm{Re} \leq 100$ the forces do not see the difference between the two conditions at the accuracy of this study. This also removes the outlet as the suspected cause of the small Δp offset in 2D-2.

Table 6: DFG 2D-2, $\Delta t = 1\,\mathrm{ms}$: errors [%] against FEATFLOW level 6 with our default outlet condition (p) and with the parabola prescribed at the outlet.

mesh	outlet	$C_{D,\max}$	$C_{D,\min}$	$C_{L,\max}$	$C_{L,\min}$	St	Δp
U4	p	-0.104	-0.099	-0.188	$+0.260$	$+0.046$	$+0.175$
U4	parabola	-0.104	-0.099	-0.189	$+0.260$	$+0.046$	$+0.175$
A1	p	-0.203	-0.196	-0.308	$+0.350$	-0.059	$+0.060$
A1	parabola	-0.203	-0.196	-0.308	$+0.350$	-0.059	$+0.060$

5. Diagnosis of remeshing-induced force oscillations

Section 4 established the accuracy of the complete solver. This section asks which parts of it control the force errors, by switching one choice of Section 3 at a time on configuration A1 (DFG 2D-2, $\Delta t = 1\,\mathrm{ms}$). Two indicators are used. The force-oscillation indicator

$$\mathcal{G} = \max_{n:\, t^n \geq 6.5\,\mathrm{s}} \left| C_D^{n+1} - 2C_D^n + C_D^{n-1} \right|$$

is the largest temporal second difference of the drag, not a step-to-step change. On the fixed mesh U3 it is 4.4×10^{-5}, the curvature of the smooth shedding signal at $\Delta t = 1\,\mathrm{ms}$. A global maximum does not by itself say where the excursion occurs, so for the re-projection runs we also split it, using the node and element counts of the force history: $\mathcal{G}_\mathrm{r}$ is the maximum over the steps whose second difference involves the first step m on a new mesh or the two steps after it ($m - 1 \leq n \leq m + 2$), and $\mathcal{G}_\mathrm{o}$ the maximum over all other steps, so that $\mathcal{G} = \max(\mathcal{G}_\mathrm{r}, \mathcal{G}_\mathrm{o})$. In every adaptive run of this section $\mathcal{G} = \mathcal{G}_\mathrm{r}$. The wall-pressure oscillation is the mean node-to-node oscillation of the pressure on the cylinder, $|p_i - (p_{i-1} + p_{i+1})/2|$, at $t = 8\,\mathrm{s}$.

5.1. Re-projection after a remesh

Our re-projection differs from the simplest choice, the unstabilised projection (12) of $\mathbf{u}^n$, in two ways: it treats both BDF2 levels, and it uses the scheme's stabilised pressure step. Table 9 switches the two independently.

Table 7: Ablations on A1 (DFG 2D-2, $\Delta t = 1\,\mathrm{ms}$): errors [%] against FEATFLOW level 6, force-oscillation indicator $\mathcal{G}$ and wall-pressure oscillation.

variant	$C_{D,\max}$	$C_{D,\min}$	$C_{L,\max}$	St	Δp	$\mathcal{G}$	wall osc.	wall [s]
A1 (all choices)	−0.20	−0.20	−0.31	−0.06	+0.06	7.7×10^{-5}	0.004	64
no keep radius	−0.30	−0.27	−0.94	−0.08	−0.04	3.4×10^{-3}	0.005	64
no minimum level	−0.60	−0.58	−0.87	−0.27	−0.03	7.7×10^{-5}	0.004	66
OSS factor 0.03	−0.22	−0.21	−0.33	−0.05	+2.07	8.0×10^{-5}	0.034	–
rotational term from $\mathbf{u}^*$	−0.22	−0.21	−0.34	−0.05	+0.57	1.5×10^{-4}	0.035	60
SUPG $T = 0.5$ ms	−0.21	−0.20	−0.34	−0.06	+0.05	7.5×10^{-5}	0.004	62
SUPG $T = 2$ ms	−0.19	−0.19	−0.27	−0.06	+0.07	8.5×10^{-5}	0.004	62
no overlap ($K = 0$)	−0.20	−0.20	−0.30	−0.06	+0.06	9.2×10^{-5}	0.004	69

The projection operator is what matters. With the unstabilised projection, $\mathcal{G}$ is 2.9×10^{-2} for $\mathbf{u}^n$ alone and 2.3×10^{-2} for both levels, 300 to 400 times the 7.7×10^{-5} of A1. The excursion is the same at every remesh: over the 150 remeshes with $t \geq 6.5\,\mathrm{s}$, the largest second difference near each remesh lies between 2.87 and 2.94×10^{-2} ($\mathbf{u}^n$ alone), and between 2.23 and 2.28×10^{-2} (both levels). At the other steps the excursion has decayed but not vanished: $\mathcal{G}_\mathrm{o}$ is 9.4 and 8.2×10^{-4}, about 20 times the fixed-mesh value. With the scheme's step, $\mathcal{G}_\mathrm{o}$ is 4.6×10^{-5}, the fixed-mesh value of U3, and the largest second difference near a remesh is at most 7.7×10^{-5} (median over the remeshes 5.1×10^{-5}). The spikes of the unstabilised projection (Fig. 6) point downwards and corrupt the extrema: $C_{D,\min}$ is −0.61% or −0.51% instead of −0.20%, and the apparently better $C_{D,\max}$ is an artefact of the same spikes. With the scheme's step the spikes disappear whether one level or both are re-projected; re-projecting $\mathbf{u}^{n-1}$ as well halves the remaining $\mathcal{G}$ (1.4×10^{-4} to 7.7×10^{-5}) and leaves the statistics unchanged to 0.003 percentage points. Not re-projecting at all is better than the unstabilised projection: $\mathcal{G}$ is 6.1×10^{-4}, eight times that of A1 but 40 to 50 times smaller than with the unstabilised projection, and the statistics move by at most 0.03 percentage points (St). Adding the next-step term of Braack et al. [8] (last row) changes nothing measurable: $\mathcal{G}$ is 7.5×10^{-5} and the statistics move by at most 0.001 percentage points. The history level and the projection operator thus have separable effects, and the operator dominates: the time step leaves $\mathbf{u}^n$ with the stabilised divergence of (9), the unstabilised projection removes it, and the next pressure step sees an artificial change of divergence at every remesh. An inconsistency between the two BDF2 levels, the other plausible cause, is not the main one: making the levels consistent with the unstabilised

projection reduces $\mathcal{G}$ by only a fifth.

Table 8 isolates the operator from the transfer: the same re-projections are applied every 10 steps on the unchanged U3 mesh, so that an exact identity would change nothing. With the scheme's step the velocity changes by at most 1.8×10^{-5} per re-projection (the velocity scale is 2), $\mathcal{G}$ rises from 4.4×10^{-5} to 9.0×10^{-5}, and the statistics change by at most 0.0003 percentage points; the next-step term again makes no difference. The unstabilised projection changes the velocity by 1.9×10^{-3}, a hundred times more, raises $\mathcal{G}$ 40- to 50-fold and moves the statistics by up to 0.013 percentage points, on a mesh that does not change. The same test on the manufactured solution (U1–U5, re-projection every 10 steps) leaves all errors of Table 2 unchanged to three significant digits.

Table 8: Re-projection every 10 steps on the unchanged mesh U3 (DFG 2D-2, $\Delta t = 1\,\mathrm{ms}$): largest change of $\mathbf{u}^n$ per re-projection (mean over $t \geq 6.5\,\mathrm{s}$), $\mathcal{G}$, and errors [%] against FEATFLOW level 6. The first row is the plain U3 run.

re-projected	with	$\max \lvert\mathbf{u}_{\mathrm{new}} - \mathbf{u}\rvert$	$\mathcal{G}$	$C_{D,\max}$	$C_{D,\min}$	$C_{L,\max}$	St	Δp
none (U3)	–	–	4.4×10^{-5}	−0.511	−0.458	−1.469	+0.093	+0.403
$\mathbf{u}^n$, $\mathbf{u}^{n-1}$	scheme step	1.8×10^{-5}	9.0×10^{-5}	−0.511	−0.458	−1.469	+0.093	+0.403
$\mathbf{u}^n$, $\mathbf{u}^{n-1}$	scheme step + next-step term	1.8×10^{-5}	9.0×10^{-5}	−0.511	−0.458	−1.469	+0.093	+0.403
$\mathbf{u}^n$	scheme step	1.8×10^{-5}	1.1×10^{-4}	−0.511	−0.458	−1.469	+0.093	+0.403
$\mathbf{u}^n$	unstabilised	1.9×10^{-3}	2.2×10^{-3}	−0.503	−0.472	−1.458	+0.086	+0.411
$\mathbf{u}^n$, $\mathbf{u}^{n-1}$	unstabilised	1.9×10^{-3}	1.7×10^{-3}	−0.504	−0.465	−1.469	+0.087	+0.408

Table 9: Re-projection after a remesh on A1 (DFG 2D-2, $\Delta t = 1\,\mathrm{ms}$): which velocity levels are re-projected and with which operator (scheme step: (10); unstabilised: (12)). Errors [%] against FEATFLOW level 6, and the force-oscillation indicator near a remesh, $\mathcal{G}_{\mathrm{r}}$, and at all other steps, $\mathcal{G}_{\mathrm{o}}$ ($t \geq 6.5\,\mathrm{s}$, 150 remeshes).

re-projected	with	$C_{D,\max}$	$C_{D,\min}$	$C_{L,\max}$	St	Δp	$\mathcal{G}_{\mathrm{r}}$	$\mathcal{G}_{\mathrm{o}}$
none	–	−0.20	−0.20	−0.31	−0.03	+0.06	6.1×10^{-4}	6.1×10^{-5}
$\mathbf{u}^n$	unstabilised	−0.09	−0.61	−0.26	+0.15	+0.07	2.9×10^{-2}	9.4×10^{-4}
$\mathbf{u}^n$, $\mathbf{u}^{n-1}$	unstabilised	−0.10	−0.51	−0.26	+0.13	+0.07	2.3×10^{-2}	8.2×10^{-4}
$\mathbf{u}^n$	scheme step	−0.20	−0.20	−0.30	−0.06	+0.06	1.4×10^{-4}	4.6×10^{-5}
$\mathbf{u}^n$, $\mathbf{u}^{n-1}$ (A1)	**scheme step**	−0.20	−0.20	−0.31	−0.06	+0.06	7.7×10^{-5}	4.6×10^{-5}
$\mathbf{u}^n$, $\mathbf{u}^{n-1}$	scheme step + next-step term	−0.20	−0.20	−0.31	−0.06	+0.06	7.5×10^{-5}	4.6×10^{-5}

5.2. Mesh near the body, in the approach flow, and adaptation interval

Keep radius. If the elements near the cylinder may be coarsened, the mesh around the body changes at every remesh. $\mathcal{G}$ grows 44 times, and $C_{L,\max}$ worsens from −0.31% to −0.94%.

Minimum level. Without the floor the approach flow stays on the base mesh. The forces then carry a bias that no refinement near the cylinder removes: $C_{D,\max}$ −0.60%, $C_{L,\max}$ −0.87% and St −0.27%, about three times the drag and lift errors of A1 and four times its St error. The run costs the same, so the minimum level improved the force accuracy without a measurable increase in wall time.

Adaptation interval. With the tolerances of A1 fixed and the mesh adapted every 5, 10 or 20 ms (overlap $K = 2$, 5 and 10 steps, look-ahead over one interval), $\mathcal{G}$ is 7.8×10^{-5}, 7.7×10^{-5} and 7.9×10^{-5}: the size of the remeshing excursions does not depend on how often the mesh changes. The accuracy does, because the refined region follows the shed vortices more closely when it is updated more often: $C_{D,\max}$ is −0.16%, −0.20% and −0.22% off and $C_{L,\max}$ −0.23%, −0.31% and −0.35%, at nearly the same mean node count (51 900, 50 800 and 51 700). The cost goes the other way: the time spent in remeshing on the critical path is 35, 18 and 9 s and the wall time 80, 64 and 52 s.

5.3. Pressure stabilisation and the rotational term

OSS factor. With $c_{\text{oss}} = 0.03$ the forces hardly change, but Δp is +2.07% off. The wall pressure carries a node-to-node mode eight times the oscillation of the reference run (Fig. 7).

Rotational term. With d^* taken as the full discrete divergence $M_L^{-1}D\mathbf{u}^*$, which retains the stabilisation contribution (5), the benchmark shows the same wall-pressure mode as the small OSS factor, a Δp error of +0.57%, and twice the $\mathcal{G}$ of A1. Together with the growth of the manufactured-solution pressure error under time-step refinement (Section 3.2), this is the evidence for omitting the contribution in the present discretisation; the eigenvalue argument given there is an interpretation, not a stability result for the coupled scheme.

5.4. SUPG time scale and overlapped remeshing

SUPG time scale. Over $T = 0.5$–2 ms every quantity changes by at most 0.04 percentage points.

Overlap. Remeshing on the critical path instead of overlapping it changes the errors by at most 0.015 percentage points and costs 7% more wall time on A1. On A3 the overlap cut the wall time from 963 s to 736 s.

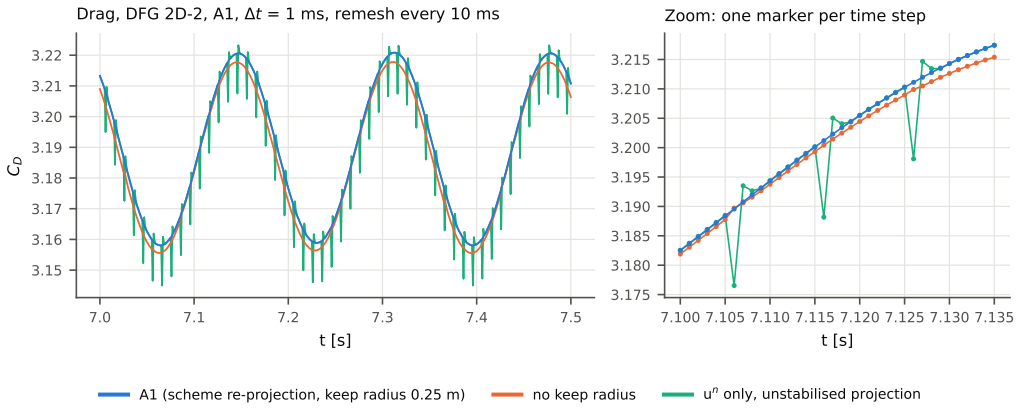


Figure 6: DFG 2D-2, A1: drag with the scheme-consistent re-projection of both BDF2 levels and the keep radius (A1), without the keep radius, and with the unstabilised projection (12) of $\mathbf{u}^n$ alone. The mesh is adapted every 10 ms (10 steps).

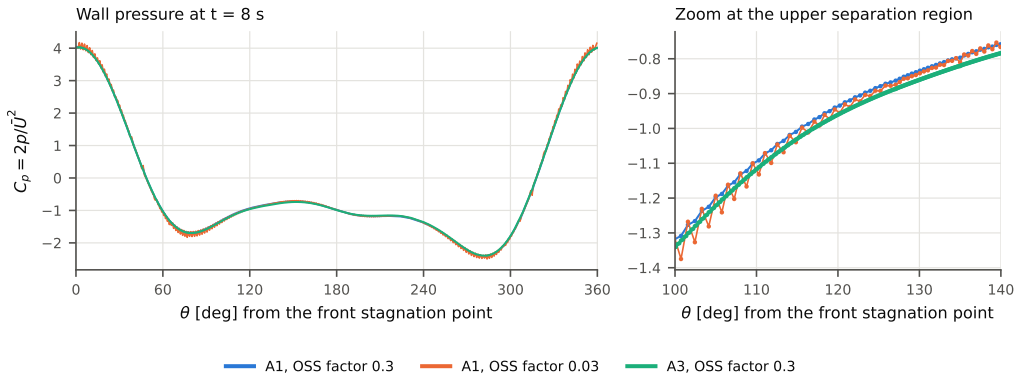


Figure 7: DFG 2D-2: pressure on the cylinder at $t = 8$ s for A1 with OSS factor 0.3 and 0.03, and for A3; θ is measured from the front stagnation point, through the upper side. The pressure level is set by the outlet condition (Section 2): the outlet pressure is close to, but not exactly, zero.

6. Cost

Fig. 3 plots the errors of the runs of Table 3 against wall time. Per node, the adaptive meshes are ahead only at the coarse end. Interpolating the uniform series at second order, A1 has about 1.5 times smaller drag error than a uniform mesh with the same number of nodes and A2 about 1.2 times; A3 has more elements than U5 (1.41M against 0.92M) and no accuracy gain. In wall time the adaptive meshes are not ahead. A1 and U4 take the same time (64 and 63 s), and A1 has twice the drag error (-0.20% against -0.10%). A3 takes 736 s for a $C_{D,\max}$ error of -0.006%, which U5 reaches in 259 s, so A3 takes 2.8 times as long for the same drag error. Each wall time is a single run; differences of a few per cent, such as 63 against 64 s, are within the run-to-run variation and are not interpreted.

Three effects contribute to the gap (Table 10). First, the adaptive runs remesh 800 times. Even with the overlap, the exposed AMR work is 29% of the A3 wall time: indicators, upload, ILU colouring, the AMG setup for the new matrix and the re-projection solve. Second, the solver costs more per node on adaptive meshes: BiCGSTAB needs 28 momentum iterations per step on A3 against 14 on U5 (Table 10). A3 has elements one level finer than U5 and transitions between refinement levels, and at a fixed Δt the finer elements have larger Courant numbers. We attribute the higher iteration count to this mesh-dependent conditioning of the momentum system, but we have not isolated the Courant number, the grading or the element shape experimentally. Third, the tolerances are uniform in space, so the far wake is resolved as finely as the boundary layer. In the A3 mesh, 480k of the 1.4M elements are at the finest level downstream of $x = 0.35$ m, although the forces hardly depend on that region. A distance-graded tolerance removed 40% of the nodes in 2D-2 at little cost in accuracy, but in 2D-3 it worsened $C_{L,\max}$ by 0.1 percentage points and brought back remesh-induced force excursions. A goal-oriented indicator [35, 36] is the principled fix, and we leave it to future work.

Table 10: Wall-time breakdown of the DFG 2D-2 A3 and U5 runs [s].

phase	A3	U5
momentum predictor (assembly, ILU(0), BiCGSTAB)	203	66
pressure (assembly, AMG setup, CG)	317	191
AMR on the critical path	213	0
indicators + download	9	
upload	8	
ILU colouring	15	
re-projection AMG setup	104	
re-projection solve	66	
overlap: transfer	9	
overlap: join wait	0	
other (forces, statistics, output)	3	2
wall time	736	259
host remesh in the worker thread (hidden)	216	
mean BiCGSTAB / CG iterations per step	28 / 38	14 / 31

7. Limitations

- Δp **in 2D-2.** Our Δp stays +0.05 to +0.22% above FEATFLOW and does not converge monotonically. The probe evaluation is not the cause: replacing the boundary-edge interpolation by local quadratic fits over 5–17 surface nodes changes Δp by 0.4% on U3 and by 0.03% on A3, a change that shrinks with refinement and changes sign between meshes. Nor is the outlet condition: prescribing the parabola there changes Δp by less than 10^{-4} percentage points (Section 4.5), although we did not implement the do-nothing condition of FEATFLOW itself. The time error is +0.02 to +0.03% (Table 4). The residual discrepancy remains unresolved.

- **Mesh-history sensitivity.** Round-off-level changes, such as a different solver or the overlapped remesh, flip individual refine and coarsen decisions. The meshes then differ at about half the remesh steps, with node counts within 0.1%. C_D and C_L change by less than 6×10^{-5}, but the 2D-3 $\Delta p(8\,\mathrm{s})$ on adaptive meshes moves by up to $\pm 0.2\%$. We do not quote it more precisely than that. The overlapped remesh, which changes the mesh history more strongly, moved the 2D-3 $C_{L,\max}$ on A2

at $\Delta t = 1\,\mathrm{ms}$ from +0.20% to +0.07%; on A3 it moved by only 0.007 percentage points. At $\Delta t = 1\,\mathrm{ms}$ this quantity carries a time error of about +0.4%, so we read the A2 change as mesh-history noise on top of that error, not as a systematic effect.

- **Wall time.** See Section 6. The per-node advantage of the adaptive meshes does not translate into a wall-time advantage at the accuracy of the reference data on this hardware.

- **Re-projection.** Our re-projection approximates the stabilised projection of Braack et al. [8] with a lumped mass matrix and the pressure-Poisson operator K in place of the Schur complement. It is not exactly the identity on an unchanged mesh, and we have no stability bound for it; the evidence is the measurements of Section 5. The transfer itself is nodal interpolation and injection, neither conservative nor divergence-preserving.

- **Scope.** Two-dimensional, laminar, one geometry. The indicators subtract the known inflow profile, which is specific to channel flows; for other flows the undisturbed state would have to be supplied or the plain recovered gradient used.

8. Conclusions

Verification. We verified a GPU P1–P1 finite-element projection solver with mesh adaptation every 10 ms on a manufactured solution and the DFG cylinder benchmarks. At $\Delta t = 1\,\mathrm{ms}$ the finest meshes reproduce the 2D-2 maximum drag to within 0.01% and the 2D-3 maximum drag to within 0.03%. The 2D-2 maximum lift is 0.11–0.19% high, which is its time error at that step; this time error is nearly the same on a uniform and an adaptive mesh, and removing it brings the extrapolated drag and lift within 0.05% of the reference, an estimate that assumes this mesh independence. The outlet condition changes the forces by less than 0.001 percentage points on uniform meshes, and stopping at $t = 8\,\mathrm{s}$ biases the lift statistics by up to 0.004 percentage points on the two meshes run to $t = 30\,\mathrm{s}$.

Diagnosis. Five implementation choices were measured by ablation. The decisive one is to re-project the transferred velocities with the scheme's own stabilised pressure step, a projection-scheme form of the method of Braack

et al. [8]. An unstabilised projection targeting zero discrete divergence produced drag excursions at every one of the 150 remeshes examined, with a second-difference indicator 300 to 400 times that of the scheme-consistent re-projection, and was worse than no re-projection at all; with the scheme's step, the indicator at steps away from a remesh equals the fixed-mesh value. The next-step term of Braack et al. [8] made no measurable difference in our scheme, and on an unchanged mesh our approximate re-projection is close to, but not exactly, the identity. The excursions do not depend on the adaptation interval between 5 and 20 ms. The other four choices are a coarsening exclusion zone around the body, a minimum refinement level for the approach flow, an SUPG time scale independent of Δt, and a rotational pressure update that omits the stabilisation part of the discrete divergence; keeping that part made the manufactured-solution pressure error grow as Δt was refined.

Cost. Per node, the adaptive meshes are 1.2 to 1.5 times more accurate than uniform ones at the coarse end and not at the fine end. On one GPU they are not faster in wall time: the finest adaptive run takes 2.8 times as long as a uniform run of the same drag accuracy, because of remeshing, a higher solver cost per node and an indicator that is not goal-oriented. A goal-oriented, adjoint-weighted indicator and a conservative or divergence-preserving transfer are the next steps.

Code and data availability

The solver source, the run scripts with every command line (mesh configurations of Table 1, adaptation parameters, solver tolerances), the force histories and statistics of every run, and the scripts that regenerate every table and figure from them are available at `https://github.com/rudhkul/cylinder-fem-amr` under the MIT licence. The VTK solution fields (about 2 GB) are not in the repository and are available from the author on request.

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